# Experiment No. 03: Illustrate various networking devices using Packet Tracer

**LEARNING OBJECTIVE:** Students should be able to illustrate various networking devices using Packet Tracer

**TOOLS:** Cisco Packet Tracer, Google and Google Docs

**THEORY:**

Cisco Packet Tracer is a powerful network simulation tool used to design, configure, and troubleshoot network topologies. It supports various networking devices like routers, switches, hubs, access points, and end devices such as PCs and servers. The tool allows users to visualize data flow and interactions between devices, making it ideal for learning and teaching networking concepts. Packet Tracer also enables users to practice configuring devices with real-world protocols, enhancing their hands-on skills in network management.

**Advantages:**

**1. Interactive Learning Environment:** Cisco Packet Tracer provides a hands-on experience for students and professionals, allowing them to design, configure, and troubleshoot network scenarios without needing physical hardware. This interactive approach enhances understanding and retention of networking concepts.
**2. Cost-Effective Training:** As a free tool for students and educators, Packet Tracer eliminates the need for expensive networking equipment. Users can simulate complex network setups and configurations, gaining valuable practical skills without the financial burden of purchasing hardware.
**3. Versatile and Comprehensive Simulation:** Packet Tracer supports a wide range of networking devices and protocols, offering a comprehensive environment for testing various network designs and configurations. Its versatility enables users to experiment with different scenarios and prepare for real-world networking challenges effectively.

**Disadvantages:**

**1. Limited Realism:** While Cisco Packet Tracer is a powerful simulation tool, it may not fully replicate the behavior of actual hardware and software in all scenarios. Some advanced features and protocols available on real devices may not be supported or accurately simulated.
**2. Learning Curve:** Despite being user-friendly, Packet Tracer can still be challenging for beginners. Users need to have a basic understanding of networking concepts to effectively utilize the tool, which might require additional learning resources or guidance.

**3. Dependency on Software Updates:** Packet Tracer's effectiveness is dependent on regular updates from Cisco to include new devices, features, and protocols. Any lag in updates can result in the tool not supporting the latest networking technologies, which may limit its usefulness in keeping up with industry trends.

## RESULT & DISCUSSION:

### ROUTERS:

Routers are essential devices in computer networks that facilitate communication between different networks by forwarding data packets from one network to another. They determine the best path for data to travel across the network based on various factors like network topology, traffic conditions, and routing protocols. Routers are typically used to connect different local area networks (LANs) or to connect a local network to a wide area network (WAN), such as the internet.




| Router Model | Name | Description | Common Applications |
|---|---|---|---|
| **4331** | Cisco 4331 ISR | Integrated Services Router with multiple services. | Branch office routers, WAN connectivity, VPNs. |
| **4321** | Cisco 4321 ISR | Smaller ISR with fewer services but robust performance. | Small businesses, remote offices, basic WAN and LAN connections. |
| **1941** | Cisco 1941 ISR | Mid-range ISR for security and communication services. | Branch offices, secure internet connections, routing, and VPNs. |
| **2011** | Cisco 2011 ISR | Compact ISR with security and WAN features. | Small to medium-sized networks, remote locations, voice and data integration. |
| **2911** | Cisco 2911 ISR | Advanced features and high performance for large networks. | Branch offices, high-performance routing, security, and VPNs. |
| **81910X** | Cisco 81910X | Mobile router with 4G LTE and other wireless options. | Mobile networks, remote connections, LTE/3G WAN applications. |

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| **819HGW** | Cisco 819HGW | High-performance mobile gateway router. | IoT, mobile applications, remote access, and wireless communication. |
| **829** | Cisco 829 Router | Compact router with IoT features. | IoT networks, smart city applications, remote locations. |
| **1240** | Cisco 1240 Series Router | Series used for small to medium branch offices. | Small office routing, WAN and LAN integration. |
| **FTRouter** | Custom/Empty Router | Likely a placeholder or custom router in a topology. | Varies depending on custom configurations and user-specific use cases. |
| **1841** | Cisco 1841 ISR | Entry-level router with routing, security, and voice. | Small businesses, WAN connectivity, routing and firewall applications. |
| **2620XM** | Cisco 2620XM ISR | Older model for branch offices with multiple services. | Small to medium offices, routing and VPNs, voice traffic. |
| **2811** | Cisco 2811 ISR | High-performance, modular ISR with extensive services. | Enterprise branch offices, high-speed WAN connections, security services. |

**SWITCHES:**

A network switch is a device that connects multiple devices (such as computers, printers, and servers) within a local area network (LAN) and uses MAC addresses to forward data to the correct destination. Switches operate primarily at Layer 2 (Data Link layer) of the OSI model, but some advanced models, known as Layer 3 switches, can also perform routing tasks.




| Switch Model | Name | Description | Common Applications |
|---|---|---|---|
| **2960** | Cisco Catalyst 2960 | A series of enterprise-grade Layer 2 switches for small to medium networks. | Used for access layer in enterprise networks, basic VLAN and Layer 2 switching. |
| **PF-Switch** | Placeholder/Special Model | Likely a placeholder or a custom switch. | Varies based on user-defined use cases or configurations. |

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| **PF-Empty** | Placeholder/Special Model | Placeholder for an empty switch in a topology. | Used for custom scenarios in network simulations. |
| **3560 24PS** | Cisco Catalyst 3560 24PS | Layer 3 switch with Power over Ethernet (PoE) capabilities, designed for high-performance environments. | Used in campus networks, with PoE for IP phones, and Layer 3 routing features. |
| **3650 24PS** | Cisco Catalyst 3650 24PS | Layer 3 switch with enhanced PoE capabilities, ideal for larger networks. | Enterprise networks, high-performance switching, VoIP, and wireless services. |
| **3650 24PS (Alt)** | Cisco Catalyst 3650 24PS (Alternative Model) | Similar to the previous model, designed for high-density environments. | Larger networks, connecting multiple devices with high-bandwidth needs. |
| **IE 2000** | Cisco Industrial Ethernet 2000 | A ruggedized switch designed for industrial environments and automation. | Used in factories, control systems, and outdoor industrial networks. |
| **PF-Bridge** | Placeholder/Special Model | Likely a placeholder for a bridge in a network topology. | Used in scenarios where bridging between network segments is required. |
| **2950-24** | Cisco Catalyst 2950 24-Port | A Layer 2 switch with 24 ports for connecting workstations and devices. | Small to medium office networks, basic switching, and network segmentation. |
| **2950T** | Cisco Catalyst 2950T | Enhanced version of the Catalyst 2950 with more advanced features. | Used for medium-sized businesses or as an access layer switch. |

**CONNECTIONS:**

In computer networking, connections refer to the physical or logical paths used to transmit data between devices. These connections can be wired or wireless and come in various types, each designed for specific uses within a network.



| Connection Type | Name | Description | Common Applications |
|---|---|---|---|
| **1st Icon (Thunderbolt)** | Serial Connection (Solid Blue) | Represents a serial connection used for direct communication between devices over long distances. | Connecting routers or other networking devices over a serial link for WAN connectivity. |
| **2nd Icon (Diagonal Line)** | Straight-Through Cable | A common Ethernet cable that connects devices like computers, routers, and switches, usually for LAN setup. | Used for connecting different devices on the same network, like a PC to a switch. |
| **3rd Icon (Dashed Line)** | Dotted Line (Virtual Link) | Represents a virtual connection, often used to signify logical connections or virtual interfaces. | Used in simulations for representing logical network connections or software interfaces. |
| **4th Icon (Blue Line)** | Ethernet Cable | A cable used for high-speed communication between devices. | Connecting PCs, switches, and other Ethernet-compatible devices in LAN setups. |
| **5th Icon (Red Zigzag)** | Cross-Over Cable | A specialized Ethernet cable used to connect similar devices (e.g., switch-to-switch or router-to-router). | Used to connect two similar devices directly, such as two switches or two computers without a hub or router. |
| **6th Icon (Curved Line)** | Fiber Optic Cable | A high-speed connection that uses light to transmit data, offering high bandwidth and long-distance capabilities. | Connecting high-performance servers, data centers, and backbone connections across long distances. |
| **7th Icon (Orange Line)** | Serial Connection (Dashed) | Used for a serial connection where the physical medium (e.g., DSL, ISDN) is used. | For connecting routers or serial devices across WANs, commonly used for slower, long-distance communication. |
| **8th Icon (Red with Circle)** | Loopback Cable | A cable used for testing the network interface on a device. | Used in troubleshooting and diagnostics to test network ports and interfaces. |

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| **9th Icon (Green Line)** | Fiber Optic Cable (Green) | Another version of fiber optic cables used in some setups, usually to represent a different type of fiber. | Used for high-speed, long-distance data transmission between servers and switches. |
| **10th Icon (Purple Line)** | Coaxial Cable | A cable used for transmitting data, less commonly used for LANs, but found in older networking setups. | Historically used in older Ethernet networks (10Base2) or cable internet setups. |

**LEARNING OUTCOME:** The student should have the ability to

**L.O1**. Enhanced Network Troubleshooting Skills: By using Cisco Packet Tracer, learners will develop the ability to identify and resolve common network issues, gaining confidence in troubleshooting complex network configurations.

**L.O2.** Proficiency in Network Configuration: Learners will acquire hands-on experience in configuring various network devices and protocols, such as routers, switches, VLANs, and routing protocols, preparing them for real-world networking tasks and certifications.

**COURSE OUTCOME:**

**C.O 1.** Competence in Network Design and Implementation: Upon completion of the course, students will be able to design, implement, and manage complex network topologies using Cisco Packet Tracer, demonstrating proficiency in configuring routers, switches, and other network devices.

**C.O 2.** Ability to Diagnose and Resolve Network Issues: Students will be equipped with the skills to diagnose and troubleshoot network problems, effectively using Cisco Packet Tracer to simulate real-world network scenarios and apply solutions to ensure network reliability and performance.

**CONCLUSION:**

**For Faculty Use**

| Correction Parameters | Formative Assessment [40%] | Timely completion of Practical [40%] | Attendance / Learning Attitude [20%] | |
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| Marks Obtained | | | | |

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